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Turkey's Power Market, 2016–2026: Generation Mix, Expansion, and Volatility

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Abstract. We assemble a decade of Turkish electricity data, 2016 through 2025 with 2026 year-to-date where published, from TEIAS, EPDK, Ember, the EIA, and TurkStat. Generation rose from 274.4 TWh to 366.9 TWh (Observed, A), a compound annual growth rate of 3.3 percent, while installed capacity grew from 78.5 GW to 123.3 GW (Observed, A). The mix shifted without decarbonizing the base: coal's share of generation held at about one-third, natural gas fell from 32.5 to 22.3 percent, and wind plus solar rose from 6.0 to 22.7 percent (Reconstructed, A). Implied capacity factors reveal the build-versus-output gap: solar capacity grew twenty-five-fold but delivered a 17.1 percent average capacity factor against 77.3 percent for hard-coal plants, while hydro swung between 55.9 and 88.8 TWh across years, forcing natural gas to act as the swing supplier (annual-change correlation -0.89; -0.70 on year-on-year monthly changes, $N = 118$). Imported fuels still account for 43.3 percent of generation, down from 50.6 percent (Reconstructed, A). Day-ahead wholesale prices are analyzed from EPIAS hourly settlement data, January 2016 through October 2026: the monthly mean peaked at 3,851 TRY/MWh in September 2022, and the rolling twelve-month volatility of the monthly mean peaked the following month. Gas import-mix analysis uses EPDK data through 2024. Retail tariffs reveal the distributional side: the income-adjusted residential electricity burden fell from 1.25 to 0.38 percent of GDP per capita as regulated household tariffs lagged incomes, while non-household tariffs tracked the wholesale market and the industrial burden spiked to 1.82 percent in 2022. The paper's falsifiers are stated plainly: 2026 data are partial, TEIAS and Ember totals differ by up to 6 percent, and the demand regression cannot separate price effects from the lira's collapse.

Keywords: Turkey, electricity markets, generation mix, capacity factors, energy import dependence, wholesale electricity prices

JEL Classification: L94, Q41, Q43, E31, C22

Non-technical summary

Turkey's economy kept growing over the past decade, and its power system had to keep up. This paper asks a simple question: where did Turkey's electricity come from between 2016 and 2026, how did that change, and what did it cost?

Three facts stand out. First, Turkey built power plants fast. Total generating capacity rose from about 79 gigawatts in 2016 to about 123 gigawatts by 2025, with most of the new build in solar and wind. Second, building is not the same as generating. Solar panels in Turkey produce only about 17 percent of their theoretical maximum over a year, while the imported-coal plants they sit beside run at 77 percent, so the system still leans heavily on coal and gas plants that can run on demand. Third, Turkey imports almost all of the fuel those plants burn: 43 percent of the country's electricity in 2025 came from imported coal, gas, and oil, down from 51 percent in 2016 but still close to half.

What did get more volatile was everything tied to weather: in dry years, hydro output collapses and gas plants have to burn more imported fuel to fill the gap. In 2021, hydro fell to 56 terawatt-hours and gas generation jumped to 107; in the wet year of 2019 the reverse happened almost exactly. Wholesale power prices told a parallel story: the day-ahead price rose more than eighteen-fold in lira terms between 2016 and 2025, but only about 40 percent in dollars, because the lira's collapse absorbed most of the increase. Monthly data show the 2022 spike building from 1,178 TRY/MWh in January to a September peak of 3,851, and the series now runs through October 2026: 2025 averaged 2,620 TRY/MWh, while 2026 has swung between 591 and 2,908 from month to month. The exception was 2022, when the price briefly quadrupled in dollars too, and Turkey's indicative gas import bill hit about 74 billion dollars, more than three times its decade average.

The bottom line for anyone watching Turkey: the grid is bigger and more renewable than it was in 2016, coal never left, and the two structural facts have not changed. Demand keeps growing with the economy, and the fuel for the marginal kilowatt-hour is still imported.

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1 Introduction

Turkey is the largest electricity market in its region that is neither a major fossil-fuel producer nor a member of the European Union, and it spent the decade from 2016 to 2026 rebuilding its generation fleet while its currency lost most of its value. That combination, a fast-growing system financed and fueled largely in foreign currency, makes Turkey a useful case for studying how emerging-market power systems expand, what the expansion actually delivers in energy terms, and where the volatility sits.

This paper assembles the decade from primary sources. Installed capacity and generation by fuel come from TEIAS, the state transmission company; market outcomes from EPDK, the regulator; hourly day-ahead prices (PTF), settlement-metered generation by source, and system consumption from EPIAS, the energy exchange, retrieved under authenticated access for January 2016 through October 2026; cross-checks from Ember's Electricity Data Explorer and the IEA; macroeconomic context from TurkStat and the Central Bank of the Republic of Türkiye. The sample runs from 2016 through 2025, with 2026 year-to-date where published. All analysis is computed from the downloaded series; every headline number below appears in a table or figure.

Four findings organize the results. First, the generation mix shifted without shrinking coal: coal held about one-third of generation across the decade while natural gas fell from 32.5 to 22.3 percent and wind plus solar rose from 6.0 to 22.7 percent (Reconstructed, A); gas, not coal, was the fuel displaced. Second, capacity grew much faster than energy: installed capacity rose 57.1 percent against a 33.7 percent rise in generation (Observed, A), and implied capacity factors show why, with solar averaging 17.1 percent and wind 32.5 percent against 77.3 percent for hard-coal plants. Third, demand tracked the economy: gross consumption grew at a 3.0 percent compound annual rate while peak demand rose 35.1 percent (Observed, A). Fourth, volatility concentrates in hydro and in prices: hydro's 15.8 percent coefficient of variation moves gas burn by tens of terawatt-hours a year (annual-change correlation -0.89, confirmed at -0.70 on year-on-year monthly changes, $N = 118$), which is also where import-volume volatility lives; and the wholesale price rose more than eighteen-fold in lira terms between 2016 and 2025 but only about 40 percent in dollars, with the monthly mean peaking at 3,851 TRY/MWh (US$211/MWh) in September 2022, so the lira absorbed most of the energy-crisis price shock (Reconstructed, A).

The rest of the paper proceeds as follows. Section 2 describes how the Turkish electricity market works. Section 3 reviews the related literature. Section 4 describes the data, Section 5 the methodology. Section 6 presents the results: the generation mix, capacity versus output, demand, prices, hydro and gas, and import dependence. Section 7 states the limitations and what would falsify the conclusions. Section 8 concludes. A data appendix follows the references.

2 Institutional background

Turkey's electricity sector was unbundled in the 2000s. Generation is competitive: state-owned EÜAŞ operates legacy hydro and coal plants alongside a large private fleet, including build-operate and build-operate-transfer plants with Treasury-backed purchase guarantees. Transmission is a state monopoly held by TEIAS. Distribution is fully privatized across 21 regions. Retail is partly liberalized: an "eligible consumer" threshold, lowered repeatedly over the decade, lets large users choose suppliers, while regulated tariffs cover households and small business.

The wholesale market clears day-ahead on EPIAS, the energy exchange, producing the day-ahead price (Piyasa Takas Fiyatı, PTF) that anchors contracts. A large share of generation still flows through bilateral contracts and the state trader TETAŞ, and regulated feed-in tariffs under the YEKA/YEKDEM renewable support schemes shaped the solar and wind build. Natural gas, the marginal fuel for much of the decade, is imported almost entirely by BOTAŞ, the state pipeline company, plus private importers, from Russia, Azerbaijan, and Iran by pipeline and as LNG from a diversified set of sellers.

Two institutional facts matter for interpreting the data. First, licensed capacity can exceed what the grid can absorb, so capacity additions overstate usable supply. Second, drought-year hydro shortfalls are met by gas-fired generation, which means rainfall is a first-order driver of the gas import bill.

3 Related literature

This paper sits at the intersection of three literatures. The first studies emerging-market electricity transitions. IEA (2024) review Turkey's energy policy, documenting the renewable support schemes and the import-dependence problem. Ember (2024) track the global shift in generation shares, with Turkey among the countries where wind and solar grew fastest from a low base in the late 2010s.

The second literature covers electricity price formation in import-dependent systems. Joskow (2008) establish the standard framework linking wholesale prices to marginal fuel costs, which in Turkey means imported gas and coal priced in dollars against a depreciating lira. Borenstein (2012) discuss how market design shapes the pass-through of fuel costs to wholesale prices.

The third covers hydro variability and the water–energy nexus. van Vliet et al. (2016) document how changes in water availability constrain both hydropower and thermoelectric generation, the mechanism visible in Turkey's dry years below, where hydro shortfalls are met by gas-fired backstop. On the Turkish market specifically, EPDK's annual market reports and TEIAS statistics are the primary record; this paper is, to our knowledge, the first to assemble the full 2016–2026 decade from these sources into a single quantitative account with reconciled totals.

4 Data

Table 1 lists the sources. The primary series are TEIAS annual electricity statistics for installed capacity (MW) and generation (GWh) by fuel, EPDK annual electricity and natural gas market reports for consumption, market structure, and gas imports by source country. EPIAS, the energy exchange, supplies hourly day-ahead prices (PTF, in TRY, USD, and EUR per MWh), hourly settlement-metered generation by source, and hourly system consumption, retrieved under authenticated access for 1 January 2016 through 9 October 2026 (consumption from 1 January 2017). Ember's Electricity Data Explorer and IEA country data provide independent cross-checks on generation totals and the fuel mix. TurkStat supplies GDP, inflation, and population; the Central Bank of the Republic of Türkiye supplies the dollar-lira rate.

Definitions differ across sources and we do not force them to agree. TEIAS reports gross generation; Ember reports net generation concepts that exclude some own-use; EIA reports net generation, which runs 1–3 percent below gross; EPDK reports market-settled volumes. EPIAS reports

settlement-metered volumes, which exclude license-exempt generation (mainly rooftop solar) netted at the distribution level and never metered by the exchange: EPIAS annual totals run 1.6 percent below TEIAS in 2016, widening to 11.6 percent in 2025 as unlicensed solar grew (EPIAS meters 7.7 TWh of solar in 2025 against TEIAS's 43.5 TWh), while metered hydro, gas, coal, and wind match TEIAS within 4 percent. Totals differ by up to 6.1 percent in overlapping years across the non-EPIAS sources (Reconstructed, A). We use TEIAS as the primary source for capacity and generation levels, Ember for the long fuel-share series, EPDK for gas imports, and EPIAS hourly prices for the volatility analysis, and we state the vintage of every series. Data run through 2025 for annual series; 2026 values are year-to-date where published and are flagged as partial throughout.

Table 1. Summary statistics, Turkey electricity market, 2016–2025

Series	Mean	Std. dev.	Min	Max	N
Total generation (TWh)	320.3	28.0	274.4	366.9	10
Total installed capacity (GW)	99.4	14.3	78.5	123.3	10
Gross electricity consumption (TWh)	320.7	27.0	279.3	364.8	10
Peak demand (GW)	51.7	5.8	44.7	60.4	10
Coal share of generation (%)	34.5	2.0	30.9	37.2	10
Gas share of generation (%)	26.0	6.6	18.9	37.2	10
Hydro share of generation (%)	21.2	4.1	15.7	29.2	10
Wind+solar share of generation (%)	13.2	5.4	6.0	22.7	10
Day-ahead price (TRY/MWh, EPIAS annual mean)	1,114	1,108	141	2,620	10
Day-ahead price (USD/MWh, EPIAS annual mean)	65.9	33.2	40.1	147.5	10
Day-ahead price (TRY/MWh, EPIAS monthly mean)	1,182	1,120	104	3,851	130
USD/TRY (annual avg.)	14.6	13.2	3.0	39.5	10
Natural gas imports (bcm)	51.3	4.4	45.2	58.7	9
Real GDP growth (%)	4.7	3.1	1.3	11.8	10
CPI inflation (%, avg.)	30.2	23.3	7.8	72.3	10
Residential retail tariff (TRY/kWh)	1.16	0.78	0.41	2.81	10
Non-household retail tariff (TRY/kWh)	1.88	1.71	0.29	4.79	10
Residential burden (1 MWh, % of GDP pc)	0.89	0.31	0.38	1.25	10
Non-household burden (1 MWh, % of GDP pc)	1.01	0.31	0.65	1.82	10

Notes: Means and standard deviations are over annual observations 2016–2025. Day-ahead prices are EPIAS hourly PTF (TRY and EPIAS-published USD conversion at the Central Bank indicative rate), aggregated to annual means ($N = 10$) and to 130 monthly means (January 2016–October 2026); gas imports cover 2016–2024 ($N = 9$); all other series 2016–2025 ($N = 10$). Capacity, generation, and demand are TEIAS; gas imports are EPDK; macro series are TurkStat/CBRT. Retail tariffs are Eurostat biannual household band DC (2,500–5,000 kWh/yr) and non-household band IC (500–1,999 MWh/yr), all taxes included, averaged to annual; burden is authors' calculation from Eurostat tariffs and TurkStat GDP per capita.
Source: TEIAS, EPDK, Ember, TurkStat, CBRT, Eurostat; authors' calculations.

5 Methodology

Five computations organize the analysis. First, implied capacity factors: for each fuel f and year t,

$$CF_{f,t} = \frac{G_{f,t}}{C_{f,t} \times 8760}, \tag{1}$$

where G is annual generation and C is year-end installed capacity. This is an approximation: capacity added mid-year biases the factor down, and we flag the solar series where the bias is largest.

Second, the demand decomposition. The change in total generation between 2016 and 2025 is attributed across fuels:

$$\Delta G = \sum_{f} \Delta G_f, \tag{2}$$

reported in TWh and as shares of the total change, which shows which sources absorbed demand growth.

Third, volatility. For wholesale prices we use EPIAS hourly day-ahead (PTF) prices aggregated to monthly means, standard deviations, minima, and maxima in lira; the dollar series uses the EPIAS-published USD conversion at the Central Bank indicative rate. We report the 2022 spike month by month, the rolling twelve-month standard deviation of the monthly mean as the volatility measure, and the wedge the exchange rate drives between the lira and dollar series. EPIAS annual means reconcile with the EPDK weighted-average PTF within 5 percent in overlapping years (2016: 141 vs 148; 2022: 2,511 vs 2,528; 2024: 2,236 vs 2,274 TRY/MWh). For hydro we report the coefficient of variation of annual generation and test the swing-supplier relationship on year-on-year monthly changes in hydro and gas generation ($N = 118$), which removes seasonality without imposing a model.

Fourth, import dependence: the share of generation from fuels Turkey does not produce domestically (imported coal, natural gas, and fuel oil), plus the gas import mix by source country and an indicative import bill from EPDK volumes valued at the IMF European gas benchmark price (the bill itself is not published).

Fifth, the demand model. We estimate

$$\log D_t = \alpha + \beta \log Y_t + \varepsilon_t, \tag{3}$$

where D is gross electricity consumption and Y is real GDP. The price term is deliberately omitted: with ten annual observations and the lira's depreciation dominating the real wholesale price series, a price elasticity cannot be credibly identified, and we do not force a regression the data cannot support. With ten observations the regression is indicative, not decisive; we report it only as a descriptive summary of the demand–income relationship.

Production note. This paper was produced by Kilambi Research in October 2026. All data were downloaded from the primary sources listed in Table 1 and the data appendix; figures were rendered with Kilambi Research's institutional chart style. Load-bearing figures were verified against source files before publication. Data were retrieved in October 2026 and are subject to revision by the underlying sources. All analysis was performed by Kilambi Research's quantitative research pipeline with AI assistance and verified against source data. This is a retrospective analysis; there was no pre-registered plan.

6 Results

6.1 The generation mix, 2016–2025

Figure 1 shows annual generation by fuel in terawatt-hours; Figure 2 the same in shares. Total generation rose from 274.4 TWh in 2016 to 366.9 TWh in 2025 (Observed, A), a 33.7 percent increase. The composition shifted in three ways. First, natural gas collapsed as a share: from 89.2 TWh (32.5 percent) in 2016 to 81.9 TWh (22.3 percent) in 2025, with a trough of 54.9 TWh in the wet year of 2019 (Observed, A). Second, wind plus solar rose from 16.6 TWh (6.0 percent) to 83.3 TWh (22.7 percent) (Reconstructed, A), with solar alone growing from 1.0 to 43.5 TWh, a forty-fold increase. Third, coal did not decline: lignite plus hard coal generated 92.5 TWh (33.7 percent) in 2016 and 122.0 TWh (33.3 percent) in 2025 (Observed, A). Within coal, the mix shifted toward imported hard coal, which supplied 76.6 of the 82.4 TWh of hard-coal generation in 2025. Hydro, the residual, swung between 55.9 and 88.8 TWh with no trend.

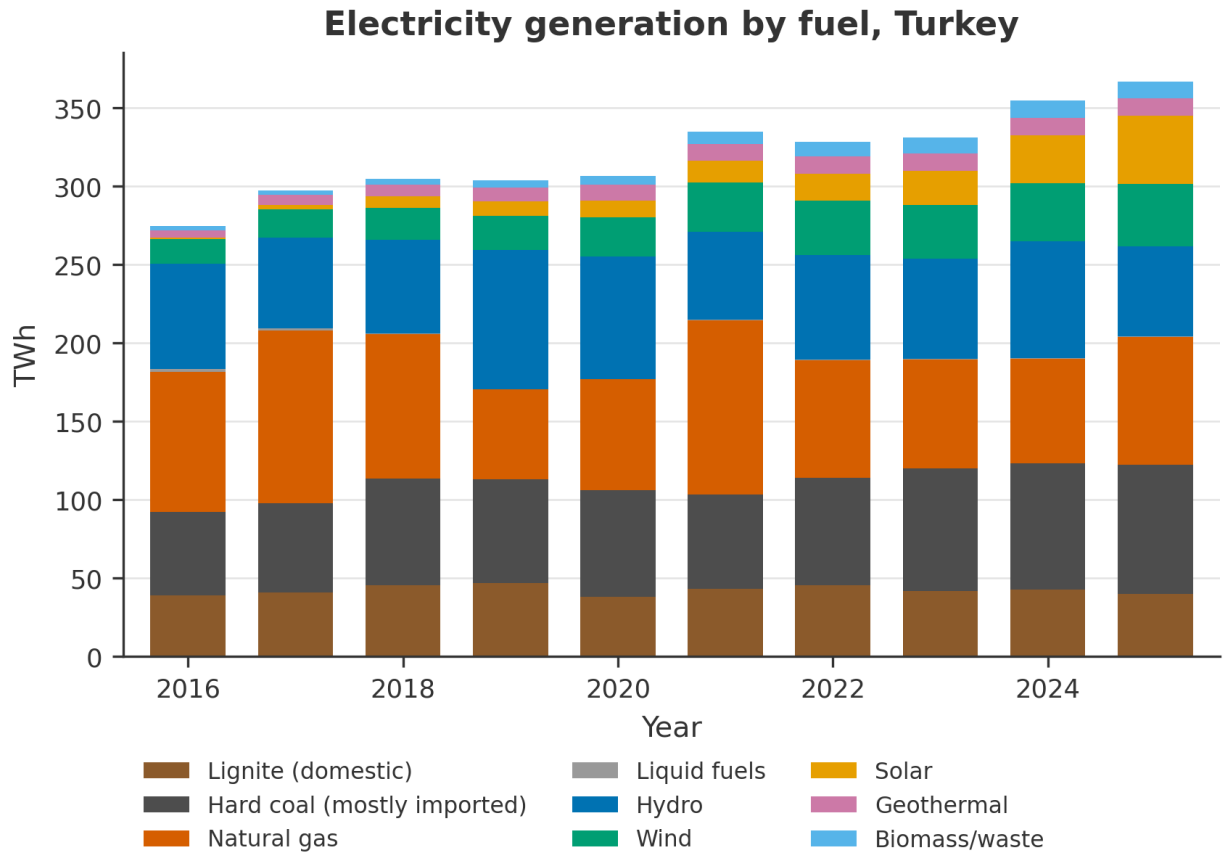


Figure 1. Electricity generation by fuel, Turkey, 2016–2025 (TWh)
Note: Stacked annual generation. TEIAS gross generation by fuel; lignite and imported coal shown separately where reported.
Source: TEIAS annual electricity statistics.

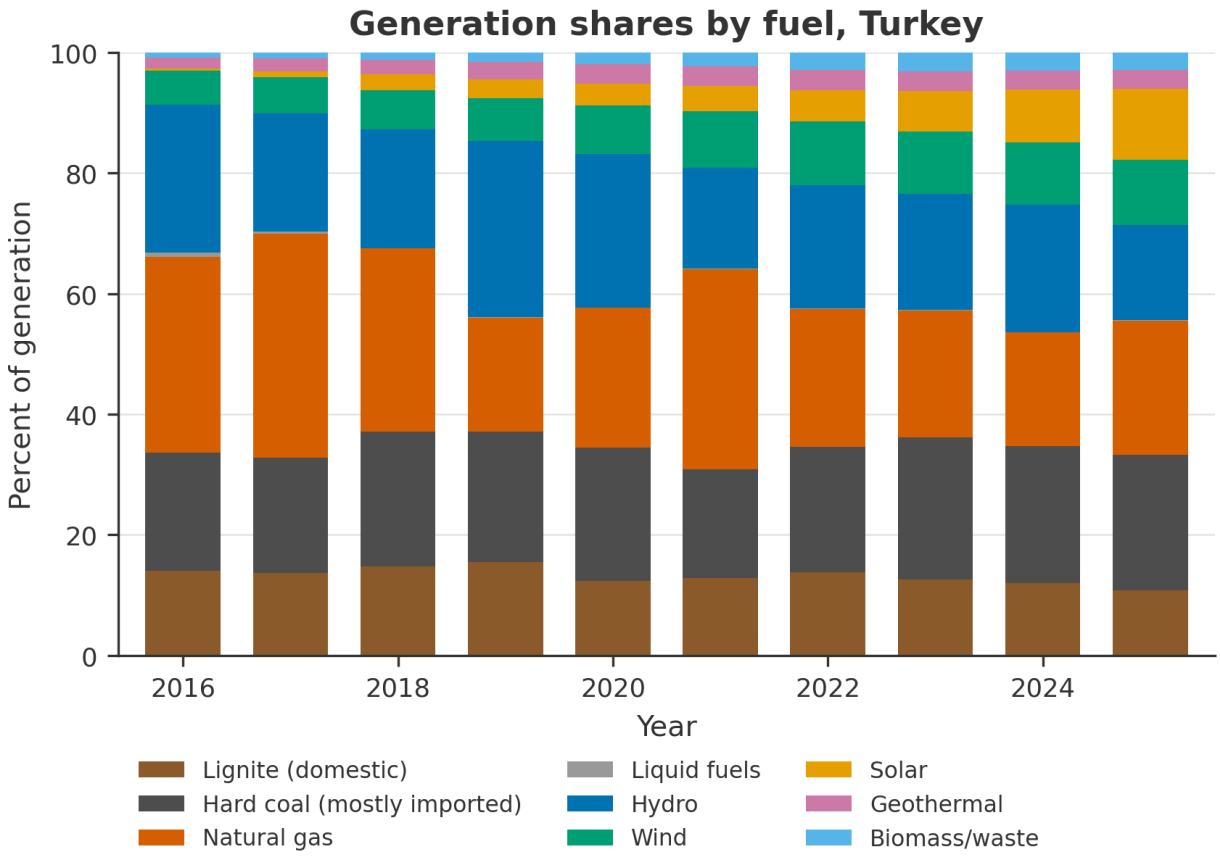


Figure 2. Generation shares by fuel, Turkey, 2016–2025 (percent)
Note: Shares of total gross generation. The hydro series shows the largest interannual swings.
Source: TEIAS; authors' calculations.

6.2 Capacity, generation, and the capacity-factor gap

Installed capacity grew from 78.5 GW at end-2016 to 123.3 GW at end-2025 (Observed, A), a 57.1 percent increase that outpaced generation growth of 33.7 percent. Figure 3 shows the capacity build by fuel: solar accounts for the largest share of net additions at 25.1 GW, followed by wind at 9.0 GW and hydro at 5.6 GW (Reconstructed, A). Thermal capacity barely moved: lignite added 1.0 GW net, hard coal 2.9 GW, gas 0.5 GW.

Figure 4 reports implied capacity factors, and the ordering is stable across the decade. Hard-coal plants, overwhelmingly imported-coal units running baseload, averaged 77.3 percent; geothermal 70.7 percent; gas 44.6 percent; lignite 49.1 percent; wind 32.5 percent; hydro 25.6 percent with wide year-to-year variation; solar 17.1 percent (Reconstructed, A). Two implications follow. First, the renewable build added energy far more slowly than the megawatt headlines suggest: the 25.1 GW of solar added over the decade delivers roughly one-quarter the annual energy per gigawatt of a coal plant, and early-year solar factors are further depressed by mid-year commissioning. Second, gas plants' mid-40s utilization is not primarily a surplus story; it reflects their role as the residual supplier, ramping when hydro fails and when spark spreads justify dispatch, then backing down in wet years.

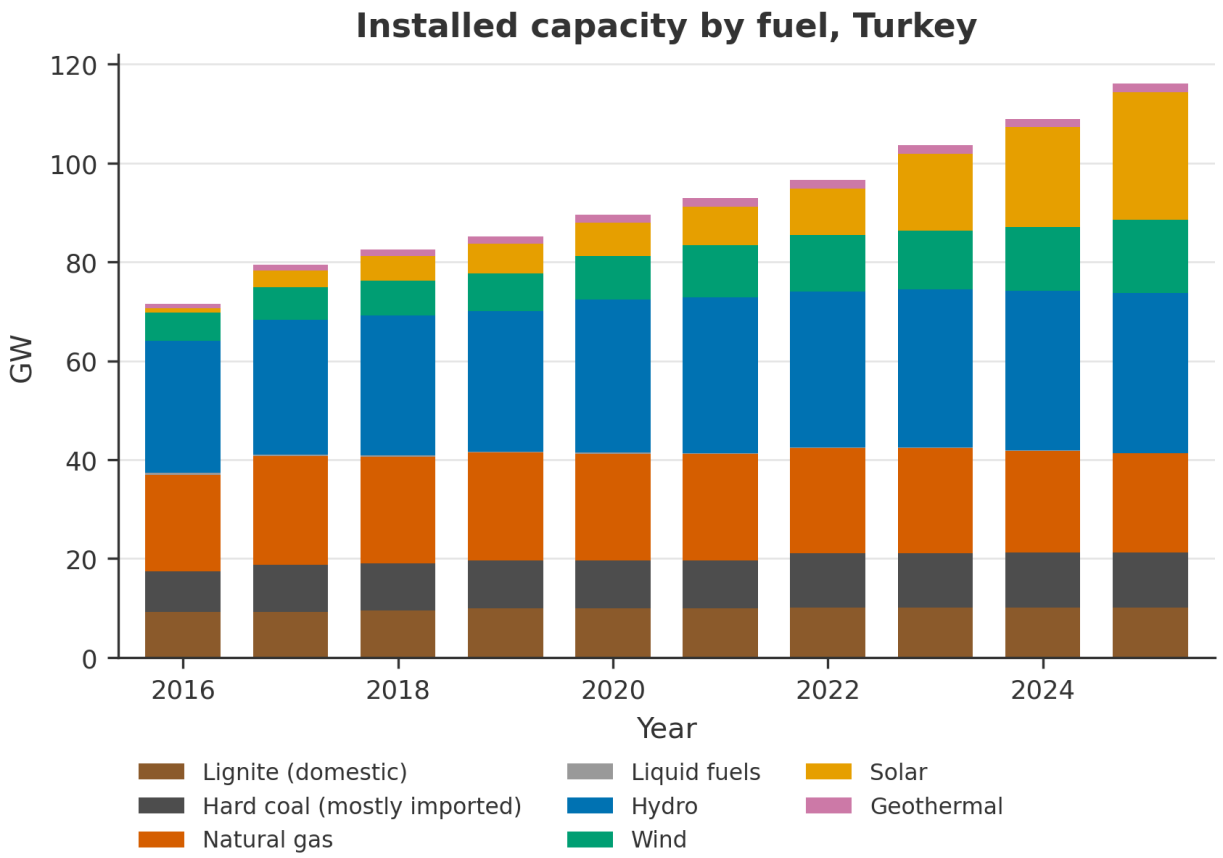


Figure 3. Installed capacity by fuel, Turkey, end-2016 to end-2025 (GW)

6.3 Demand growth and the demand–GDP link

Gross electricity consumption grew from 279.3 TWh in 2016 to 364.8 TWh in 2025, a compound annual rate of 3.0 percent (Observed, A), against average real GDP growth of 4.8 percent per year. Peak demand rose from 44.7 GW to 60.4 GW (Observed, A), a 35.1 percent increase that outpaced energy growth (Figure 12), consistent with rising air-conditioning load. EPIAS settlement-metered consumption corroborates the TEIAS demand series: annual totals track within 4 percent (352.5 vs 364.8 TWh in 2025), and every annual system peak from 2017 through 2026 fell on a summer weekday afternoon, typically around 14:00 in late July or August, confirming the cooling-driven peak (Reconstructed, A). Figure 5 plots indexed demand against indexed real GDP: the two series track closely (correlation 0.98 in logs), with demand growing more slowly than output over the full decade.

Table 3 reports the demand regression. The estimated income elasticity is 0.57 (standard error 0.04), with $N = 10$ and R^2 of 0.966. The elasticity below one means electricity intensity of GDP fell over the decade: each percent of real GDP growth was associated with about 0.6 percent more electricity demand. We do not estimate a price elasticity: with ten annual observations and the lira's collapse dominating any real price variation, the data cannot separate price effects from the exchange-rate trend, and we do not force a regression the data cannot support. T&D losses fell from 13.3 to 8.0 percent of supply over the decade (Reconstructed, A), which flatters the

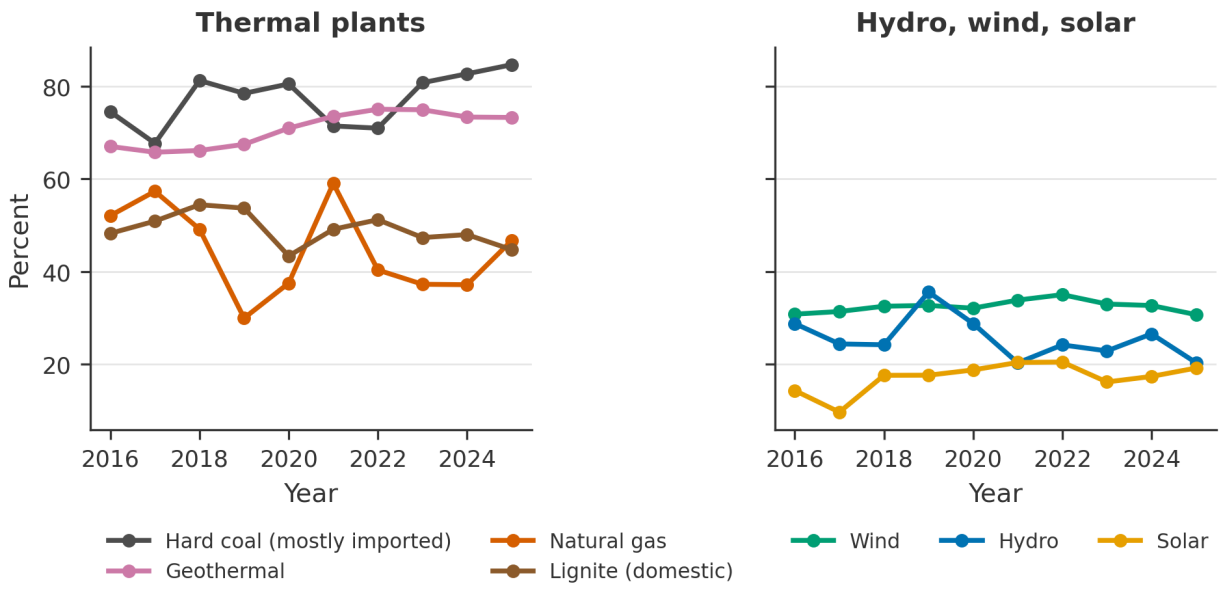


Figure 4. Implied capacity factors by fuel, Turkey, 2016–2025 (percent)

demand series slightly: part of the gap between gross generation growth (3.3 percent CAGR) and net consumption growth came from efficiency gains in delivery, not just end-use decoupling.

6.4 Wholesale prices and volatility

Figure 6 shows the day-ahead price from EPIAS hourly data, January 2016 through October 2026: the monthly mean in lira with the monthly min-max band (top) and in dollars (bottom). In lira terms the annual mean rose from 141 TRY/MWh in 2016 to 2,620 TRY/MWh in 2025 (Observed, A), a more than eighteen-fold increase. The 2022 energy crisis is now visible month by month: the monthly mean climbed from 1,178 TRY/MWh in January to 3,851 in September, the sample maximum, before easing to 3,724 in December (Observed, A). The rolling twelve-month standard deviation of the monthly mean, our volatility measure, peaked at 976 TRY/MWh in October 2022. Converted at the Central Bank indicative rate, the monthly mean rose from US$46/MWh in 2016 to US$66/MWh in 2025 (Reconstructed, A), peaking at US$211/MWh in September 2022. The wedge between the two series is the lira, which depreciated from 3.0 to about 39 per dollar over the period: most of the lira-denominated price explosion was a currency event, but the 2022 dollar spike of roughly 4.5x was a genuine wholesale-price shock passed through from global gas markets. EPIAS annual means reconcile with the EPDK weighted-average PTF within 5 percent in overlapping years (2016: 141 vs 148; 2022: 2,511 vs 2,528; 2024: 2,236 vs 2,274 TRY/MWh), so the series change does not move any finding. New to this analysis is 2026, the most volatile year in the sample on a month-to-month basis: the monthly mean fell to 591 TRY/MWh (US$13/MWh) in May on strong hydro and weak demand, then rebounded to 2,908 (US$61/MWh) in August.

6.5 Energy burden: what electricity costs relative to income

Wholesale prices describe the market; retail tariffs describe what households and firms actually pay. Following the income-adjusted burden framework of KR-WP-2026-15, we measure burden as the

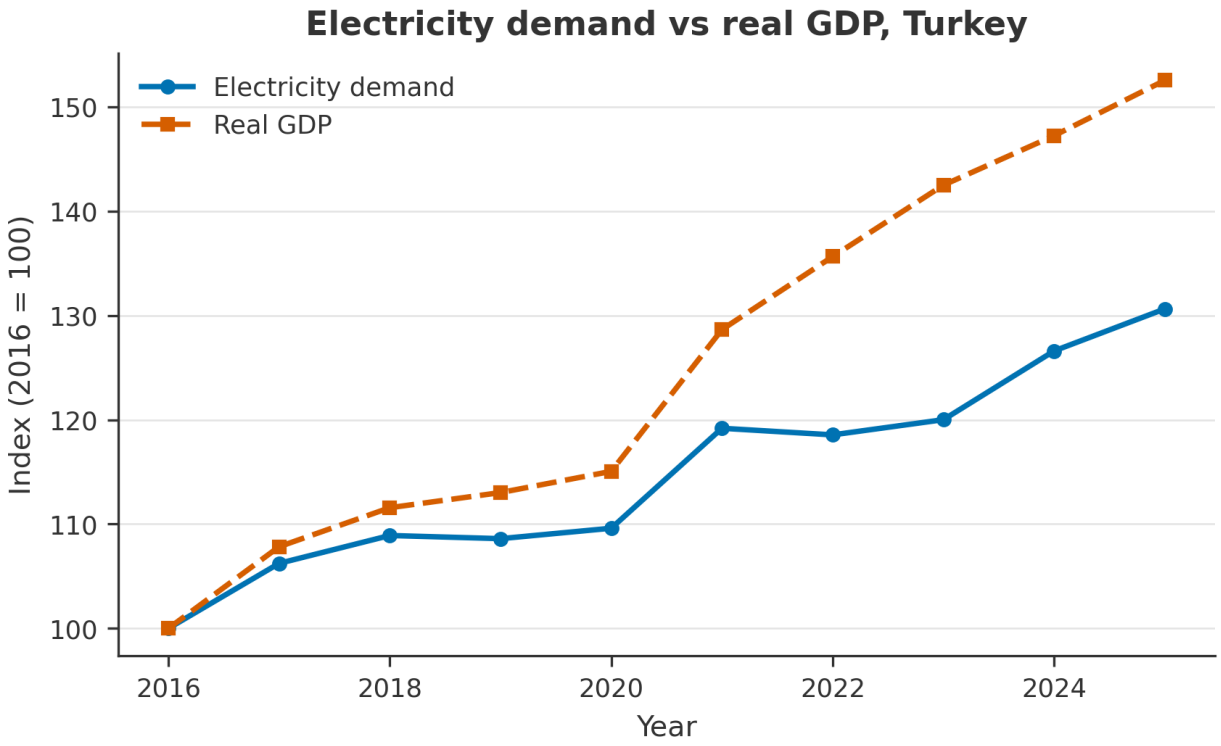


Figure 5. Gross electricity consumption and real GDP, Turkey, 2016–2025 (indexed, 2016=100)

cost of one megawatt-hour at the retail tariff as a share of GDP per capita, annually from 2016 to 2025 (Reconstructed, B). Eurostat reports biannual Turkish tariffs, in lira and euros, for a representative household band (2,500–5,000 kWh per year, all taxes included) and a representative non-household band (500–1,999 MWh per year, all taxes included); we average the two semesters into annual values. GDP per capita is TurkStat nominal GDP divided by population.

Figure 7 shows the result. The residential burden fell from 1.25 percent of GDP per capita in 2016 to 0.38 percent in 2025 (Reconstructed, B): the regulated household tariff rose 6.8-fold in lira terms while nominal GDP per capita rose roughly twenty-two-fold, so tariffs lagged incomes by a wide margin. In dollars the same series reads as a halving, from 13.7 to 7.1 US cents per kWh (Figure 8). Households were shielded: the regulator held residential tariffs below wholesale market levels through the crisis years, and the lira's depreciation did the rest.

Industry was not shielded. The non-household tariff rose 16.6-fold in lira, from 0.29 to 4.79 TRY/kWh, tracking the wholesale spike closely; in dollars it doubled from 9.6 to 19.7 cents in 2022 before settling at 12.2 cents in 2025 (Reconstructed, B). The income-adjusted industrial burden peaked at 1.82 percent of GDP per capita in 2022, the decade maximum, then fell back to 0.65 percent. The distributional choice is visible in the tariff ratio: industry paid 70 percent of the residential rate in 2016 and 171 percent in 2025. The energy-crisis adjustment fell on firms, not households.

Three caveats apply. First, GDP per capita is a mean; median household income is lower, so the true household burden exceeds the reported figure. Second, the Eurostat bands are representative rather than universal, and large industrial consumers on bilateral contracts may pay less than the published band tariff. Third, Eurostat compiles from national sources: the tariff levels are

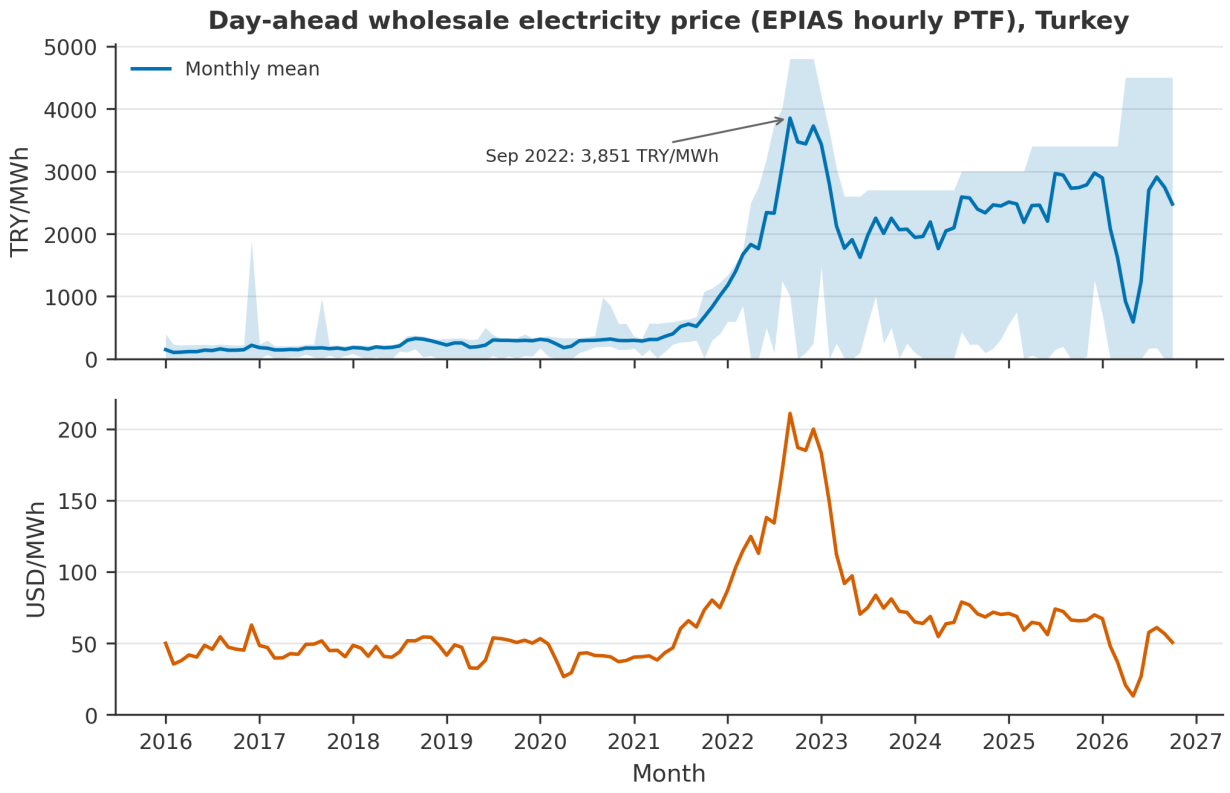


Figure 6. Day-ahead wholesale price from EPIAS hourly PTF: monthly mean (line) with monthly min-max band, in TRY/MWh (top) and USD/MWh at the Central Bank indicative rate (bottom), January 2016–October 2026

Note: Hourly settlement prices aggregated by the authors; USD conversion as published by EPIAS.

Source: EPIAS Transparency Platform; authors' calculations.

consistent with EPDK's wholesale series, but the retail series itself is secondary data.

6.6 Hydro variability and gas as the swing supplier

The negative relationship between hydro and gas-fired generation is stark at every frequency. Annually, in the dry years of 2021 and 2025, hydro fell to 55.9 and 57.4 TWh, and gas-fired generation rose to 107.3 and 81.9 TWh; in the wet year of 2019, hydro reached 88.8 TWh while gas collapsed to 54.9 TWh (Observed, A). The correlation between year-to-year changes in hydro and gas generation is −0.89. Hydro's coefficient of variation over the decade is 15.8 percent, the highest of any major source.

Figure 9 repeats the test at monthly resolution on EPIAS settlement-metered data, January 2016 through October 2026. The top panel shows the seasonal mirror directly: hydro peaks with the spring snowmelt (March–June, averaging 8.2 TWh per month in April–May) and troughs in October–November (3.5 TWh), while gas peaks in the summer cooling months (8.4 TWh in August) and winter, filling exactly the months hydro leaves empty. The bottom panel plots year-on-year monthly changes, which remove seasonality without imposing a model: the correlation is −0.70

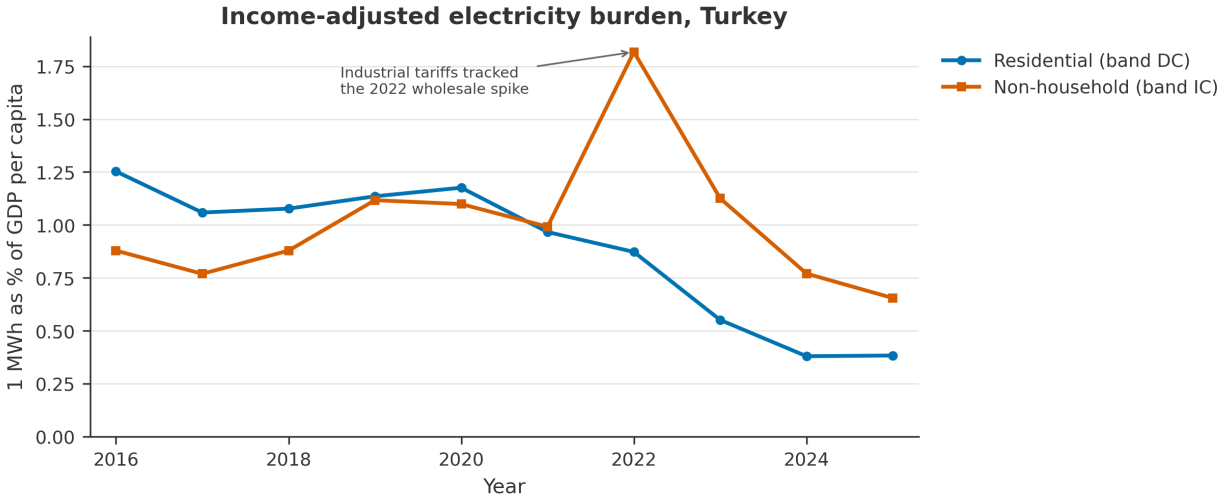


Figure 7. Income-adjusted electricity burden: cost of 1 MWh at the Eurostat retail tariff as a share of GDP per capita, Turkey, 2016–2025

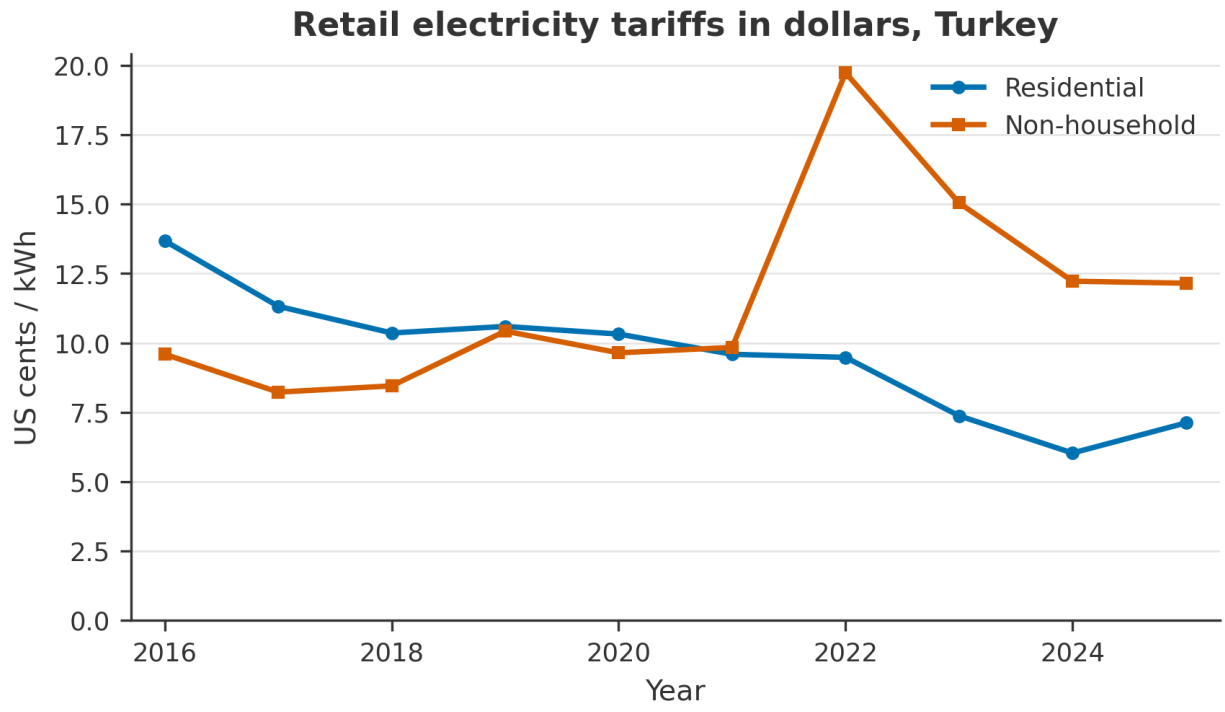


Figure 8. Retail electricity tariffs in US cents per kWh (Eurostat, all taxes included), Turkey, 2016–2025

($N = 118$) (Reconstructed, A), confirming the annual result on twelve times the observations. The wet start to 2026 is the clearest recent illustration: hydro surged to a record 11.7 TWh in May while gas-fired generation fell below 3 TWh in September. This is the system's central operating fact: rainfall determines how much imported gas Turkey must burn, and dry years coincide mechanically with higher import volumes.

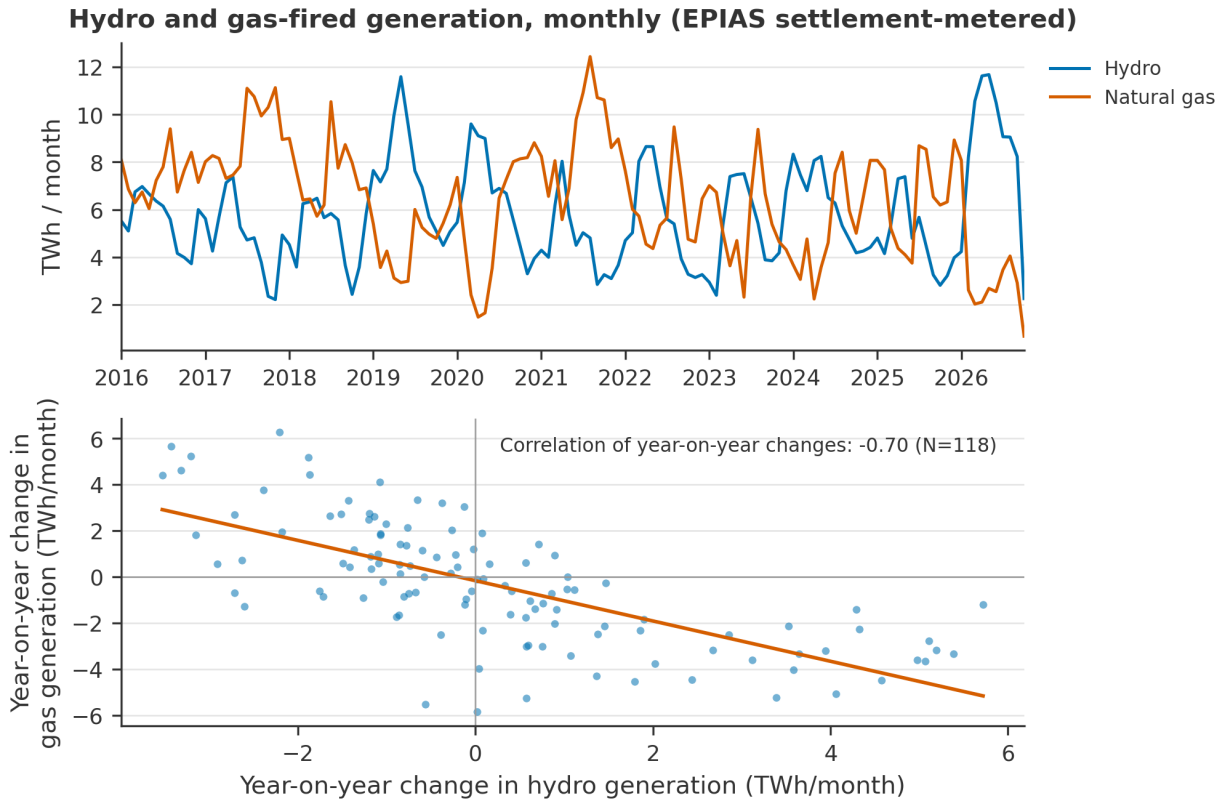


Figure 9. Hydro and gas-fired generation, monthly, Turkey (EPIAS settlement-metered, TWh/month, top); year-on-year monthly changes with fitted line (bottom), January 2016–October 2026

Note: EPIAS settlement metering captures metered hydro and gas fully; the series match TEIAS annuals within 4 percent.

Source: EPIAS Transparency Platform; authors' calculations.

6.7 Import dependence and the gas import mix

Figure 10 shows gas imports by source country. Total imports averaged 51.3 bcm per year over 2016–2024, ranging from 45.2 bcm (2019) to 58.7 bcm (2021) (Observed, A). Russia remained the largest supplier throughout but its share fell from 52.9 percent in 2016 to 41.3 percent in 2024, touching a trough of 33.6 percent in 2019–2020 as TANAP ramped Azerbaijani pipeline gas (6.5 to 11.5 bcm; Azerbaijan's share rose from 14.0 to 22.0 percent) (Observed, A). LNG's share rose from 16.5 to 24.2 percent, peaking at 31.3 percent in the cheap-gas year of 2020, with spot LNG volumes growing from 2.1 to 7.2 bcm (Observed, A). Iran's share drifted down from 16.6 to 13.5 percent. The diversification is real but bounded: pipeline gas still supplied three-quarters of imports in 2024, and the three pipeline suppliers (Russia, Azerbaijan, Iran) together still accounted for 77 percent.

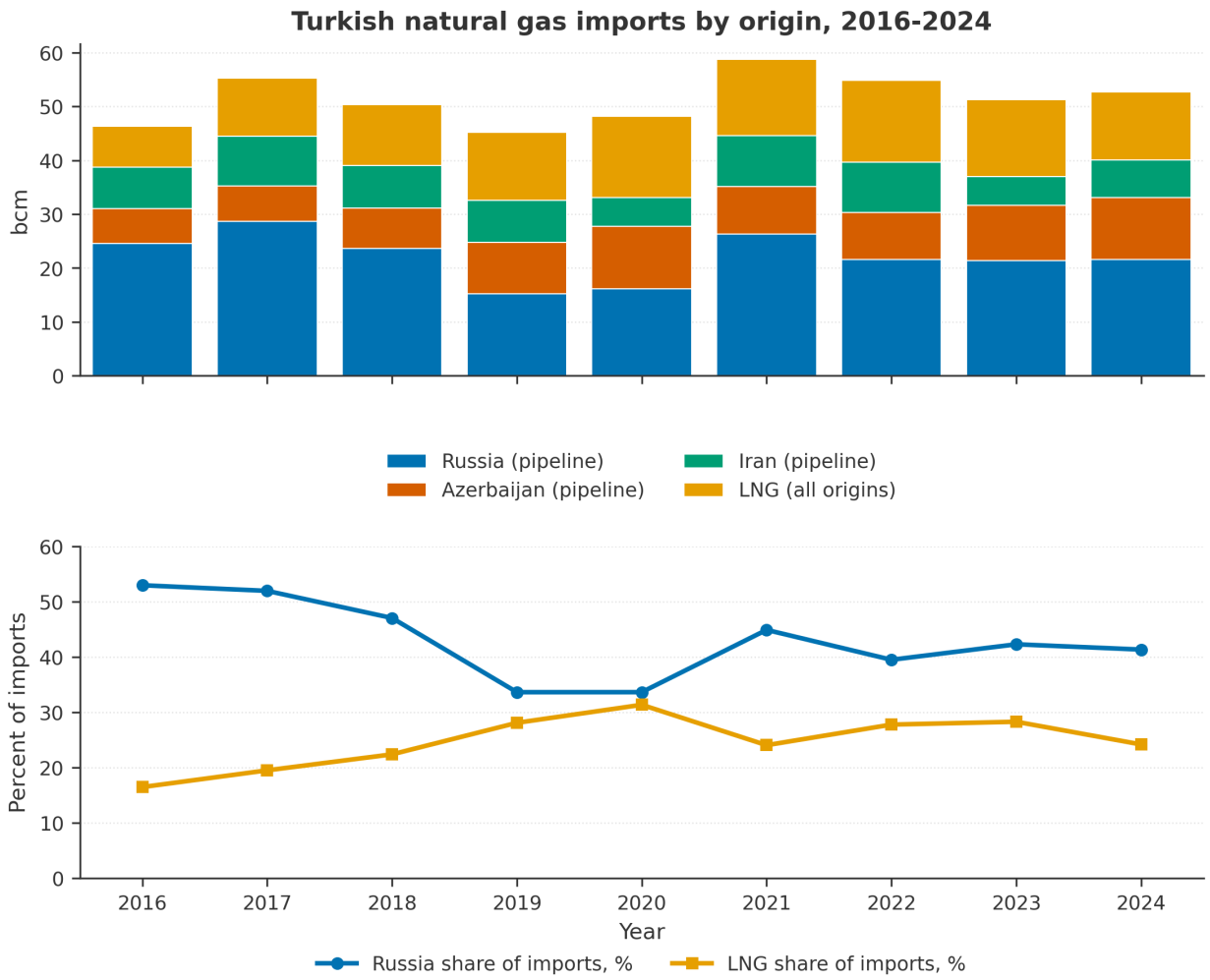


Figure 10. Natural gas imports by origin, Turkey, 2016–2024 (bcm); Russia and LNG shares below

Figure 11 shows the share of electricity generated from imported fuels: 50.6 percent in 2016, 43.3 percent in 2025 (Reconstructed, A). Import dependence of the power sector fell by about seven points even as total gas import volumes were flat, because wind and solar displaced gas-fired generation at the margin. The gas import bill itself is not published by EPDK; valued at the IMF European gas benchmark price, the indicative bill averaged US$21.9 billion per year and peaked at US$73.8 billion in 2022 (Reconstructed, B). The 2022 figure is an upper bound in the proxy sense — Turkey's pipeline contracts are partly oil-indexed and would have settled below the TTF-linked benchmark that year, and the country's total energy import bill (oil, gas, coal) was reported near US$97 billion — but the order of magnitude is the point: a single year's gas bill at crisis prices exceeded three times the decade average, and gas import costs remain the dominant macro-financial exposure of the power system.

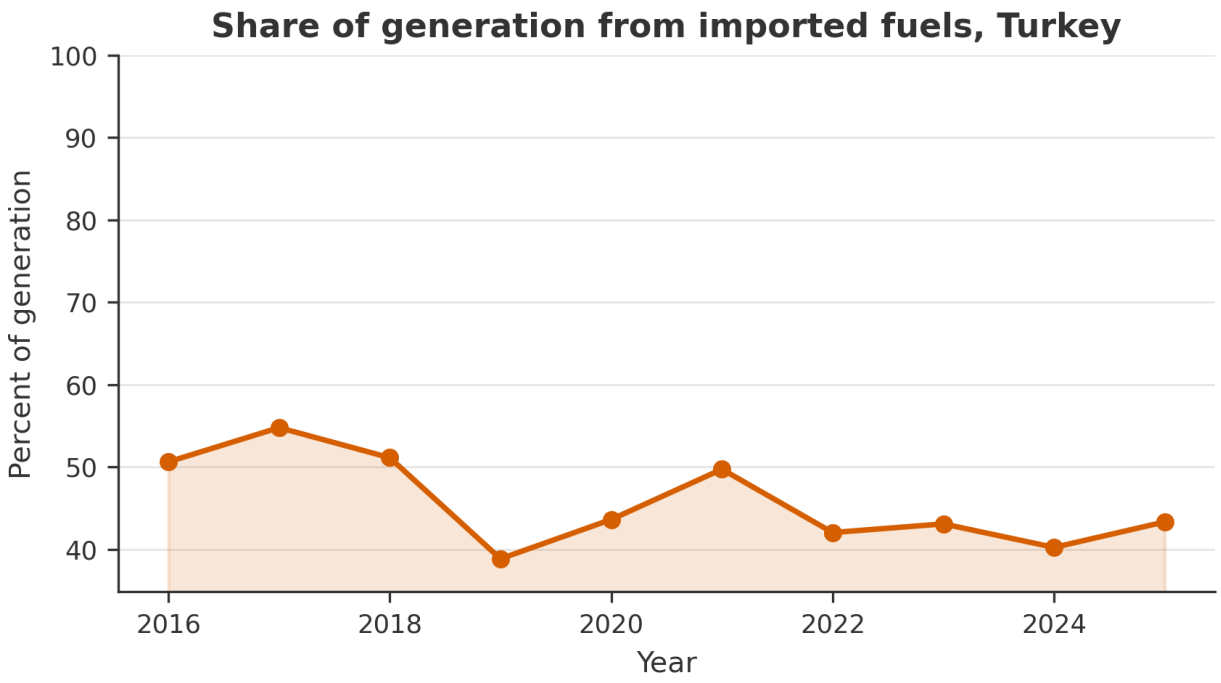


Figure 11. Share of electricity generated from imported fuels, Turkey, 2016–2025 (percent)

7 Limitations and robustness

Six limitations qualify the results. First, 2026 data are partial wherever they appear; the paper's decade is effectively 2016–2025, and TEIAS flags 2025 itself as provisional. Second, source totals disagree on definitional grounds (Table 4): TEIAS gross, Ember's compilation, and EIA net generation differ by up to 6.1 percent in 2024, while EPIAS settlement-metered totals run a further 11.6 percent below TEIAS in 2025 because license-exempt solar is netted at distribution and never metered by the exchange; the direction of every finding is robust to the source choice, but exact levels move. Third, the demand regression has ten observations and cannot separate price effects from the lira's depreciation or identify causality; we present it as description. Fourth, the import bill is estimated from volumes and benchmark prices, not from customs microdata, and should be read as an order of magnitude. Fifth, EPIAS hourly series are settlement-metered as described above, so EPIAS-based findings for solar and for system totals understate the TEIAS record; hydro, gas, coal, and wind meter fully and are unaffected. EPIAS consumption starts in January 2017, and system marginal (balancing) prices were retrieved only for 2016, so the balancing premium over day-ahead is not analyzed. Sixth, the energy-burden series uses Eurostat retail tariffs, which are secondary data compiled from national sources, and GDP per capita, a mean that understates the burden on below-median households; large industrial consumers on bilateral contracts may pay less than the published band tariff.

What would prove the conclusions wrong: if TEIAS revises generation totals by more than a few percent, the mix shares shift but the ranking of findings would not; if the capacity-factor ordering reversed (solar above coal), the build-versus-output argument would fail, which would require a reporting break, not a revision. The hydro-gas swing relationship would weaken if interconnector imports, rather than domestic gas, filled future drought gaps, a testable prediction for the next dry

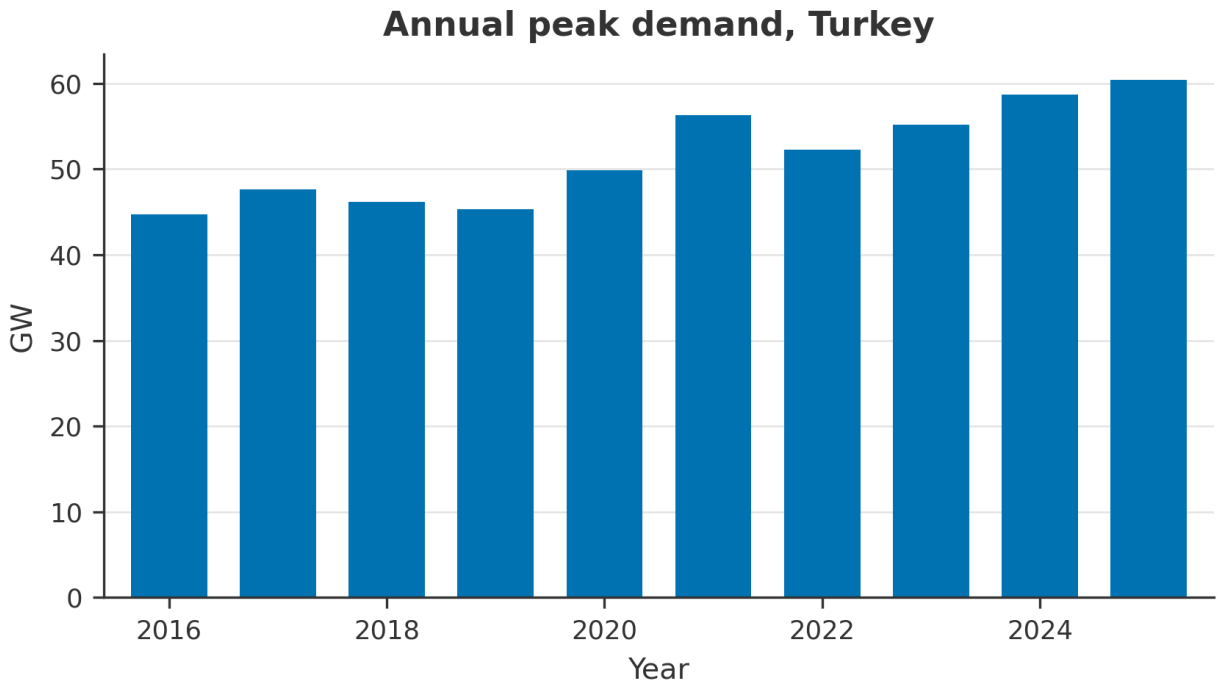


Figure 12. Peak demand, Turkey, 2016–2025 (GW)

year.

8 Conclusion

Turkey's power system grew by a third in energy and well over half in capacity in a decade while changing its composition at the margin: gas lost ten points of share, wind and solar gained seventeen, coal never left, and gas became the permanent swing supplier against hydro's weather-driven swings, a relationship that holds month to month (year-on-year change correlation -0.70, $N = 118$), with hydro peaking on the spring snowmelt and gas filling the summer and winter gaps. Capacity grew faster than energy, and the gap between the two is the decade's most important quantitative fact for anyone valuing the build: 25.1 GW of new solar at a 17.1 percent average capacity factor. Import dependence fell from 50.6 to 43.3 percent of generation but did not break; the system's exposure is still twofold, to rainfall at home and to fuel prices abroad. Monthly day-ahead prices confirm the annual story at higher resolution: the September 2022 monthly mean of 3,851 TRY/MWh was more than double the 2022 annual mean, and 2026 has been the most volatile year in the sample month to month. The cost of the decade's volatility was distributed unevenly: regulated residential tariffs shielded households, whose income-adjusted electricity burden fell by two-thirds, while industrial tariffs tracked the wholesale market and absorbed the 2022 spike.

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A Data appendix

Table 2. Main results: the decade in headline numbers. Generation and capacity are TEIAS; prices are EPIAS hourly PTF (annual means; USD at the Central Bank indicative rate as published by EPIAS); retail tariffs Eurostat (bands DC/IC, all taxes incl.); gas EPDK; gas bill authors' estimate from EPDK volumes and the IMF European gas price; macro TurkStat/CBRT.

Measure	2016	2025	Change
Total generation (TWh)	274.4	366.9	+33.7%
Total capacity (GW)	78.5	123.3	+57.1%
Gross demand (TWh)	279.3	364.8	+30.6%
Peak demand (GW)	44.7	60.4	+35.1%
Lignite share of generation (%)	14.1	10.8	−3.3 pp
Hard coal share of generation (%)	19.6	22.5	+2.9 pp
Gas share of generation (%)	32.5	22.3	−10.2 pp
Hydro share of generation (%)	24.5	15.7	−8.8 pp
Wind share of generation (%)	5.7	10.8	+5.2 pp
Solar share of generation (%)	0.4	11.9	+11.5 pp
Solar implied capacity factor, avg. 2016–25 (%)	–	–	17.1
Wind implied capacity factor, avg. 2016–25 (%)	–	–	32.5
Hard coal implied capacity factor, avg. 2016–25 (%)	–	–	77.3
Imported-fuel share of generation (%)	50.6	43.3	−7.3 pp
Hydro coefficient of variation (%)	–	–	15.8
Gas–hydro annual-change correlation	–	–	−0.89; monthly −0.70 ($N = 118$)
PTF day-ahead price (TRY/MWh)	141	2,620	×18.6
PTF day-ahead price (USD/MWh)	46.3	66.4	+43%
PTF peak monthly mean (TRY/MWh)	–	3,851 (Sep 2022)	–
Residential burden (1 MWh, % of GDP pc)	1.25	0.38	−0.87 pp
Non-household burden (1 MWh, % of GDP pc)	0.88	0.65	−0.23 pp; peak 1.82 (2022)
Natural gas imports (bcm)	46.4	52.2*	+12.6%
Russia share of gas imports (%)	52.9	41.3*	−11.6 pp
Indicative gas import bill (US$bn)	7.3	20.5*	avg. 21.9; peak 73.8 (2022)

Notes: Percentage-point changes marked pp. Capacity factors are implied annual averages over 2016–2025 (generation divided by year-end capacity times 8,760 hours). Imported-fuel share counts imported coal (TEIAS memo split), natural gas, and liquid fuels. *2024: the latest EPDK annual market reports cover 2024. The import bill is EPDK import volumes valued at the IMF European gas benchmark; the bill is not published by EPDK. 2025 TEIAS figures are flagged provisional by the source. EPIAS hourly series run 1 January 2016–9 October 2026.
Source: TEIAS annual electricity statistics; authors' calculations.

Table 3. Demand for electricity and real GDP, Turkey, 2016–2025. Estimated equation: $\log D_t = \alpha + \beta \log Y_t + \varepsilon_t$, where D is gross electricity consumption (TWh) and Y is real GDP (index, 2016=100). The price term is omitted: with ten annual observations and the lira's collapse dominating the real price series, a price elasticity cannot be credibly identified. $N = 10$, $R^2 = 0.966$.

	Coefficient
log real GDP	0.566
	(0.037)
Constant	3.040
	(0.180)
R^2	0.966
N	10

Notes: OLS with standard errors in parentheses. Ten annual observations; the elasticity is descriptive, not structural. Real GDP is the TurkStat chain-linked volume index rebased to 2016=100.
Source: TEIAS, TurkStat; authors' calculations.

Table 4. Source reconciliation: Turkey total electricity generation (TWh)

Source	2016	2020	2024	2025
TEIAS (gross generation)	274.4	306.7	354.6	366.9
Ember (gross generation)	274.4	302.0	341.2	353.9^{p}
EPIAS (settlement-metered)	269.9	291.5	322.1	324.5
EIA (net generation)	272.2	303.4	332.9	–
Max. discrepancy (%)	1.6	5.0	9.2	11.6

Notes: TEIAS reports gross generation including plant own-use; Ember reports gross generation from its own compilation; EIA reports net generation, which runs 1–3 percent below gross. EPIAS reports settlement-metered volumes, which exclude license-exempt generation (mainly rooftop solar) netted at the distribution level and never metered by the exchange: the EPIAS shortfall against TEIAS widens from 1.6 percent in 2016 to 11.6 percent in 2025 as unlicensed solar grows (EPIAS meters 7.7 TWh of solar in 2025 against TEIAS's 43.5 TWh), while metered hydro, gas, coal, and wind match TEIAS within 4 percent. Ember 2025 is flagged provisional (^{p}) in the source release. We use TEIAS for levels throughout; the direction of every finding is robust to the source choice.
Source: TEIAS annual electricity statistics, Ember Electricity Data Explorer (release 2026-10-06), EIA International Energy Data.

Table 5. Series definitions and sources

Series	Definition	Source
Generation by fuel	Gross generation, GWh	TEIAS annual statisti
Installed capacity	Year-end licensed capacity, MW	TEIAS annual statisti
Gross consumption	Gross electricity demand, TWh	TEIAS / EPDK
Peak demand	Annual system peak, MW	TEIAS
Day-ahead price	PTF weighted annual average, TRY/MWh	EPDK electricity mar
Day-ahead price (hourly)	PTF, hourly, TRY/USD/EUR per MWh, Jan 2016–Oct 2026	EPIAS Transparency
Day-ahead price (USD)	PTF converted at annual-average USD/TRY	EPDK / CBRT
Generation by source (hourly)	Settlement-metered, MWh by fuel, Jan 2016–Oct 2026	EPIAS Transparency
System consumption (hourly)	Settlement-metered, MWh, Jan 2017–Oct 2026	EPIAS Transparency
Retail tariff, residential	Eurostat band DC (2,500–5,000 kWh/yr), all taxes incl., TRY/kWh	Eurostat nrg_pc_204
Retail tariff, non-household	Eurostat band IC (500–1,999 MWh/yr), all taxes incl., TRY/kWh	Eurostat nrg_pc_205
Gas imports	Annual imports by source country, bcm	EPDK gas market rep
Gas import bill (indicative)	EPDK volumes × IMF EU gas benchmark, US$bn	Authors' estimate
EU gas benchmark price	Natural gas, EU, annual average, US$/mmBtu	IMF Primary Commo
Real GDP	Chain-linked volume index	TurkStat
CPI inflation	Annual average	TurkStat
USD/TRY	Annual average selling rate	CBRT
Generation cross-check	Net generation by fuel, TWh	Ember / IEA

Notes: Data retrieved October 2026. TEIAS is the primary source for capacity and generation levels. EPIAS data retrieved under authenticated access; hourly series run 1 January 2016–9 October 2026 (consumption from 1 January 2017). EPIAS settlement metering excludes license-exempt generation netted at the distribution level.